AGENTIC-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

THE SITE

Location	Amazon, TX -- station KSAT, America/Chicago
Committed pair	OSM 254742609 -> 1537390734 OSM way 254742609 / OSM way 1537390734
Facade gap	315.7 m between the two halls
Weather record	43,809 real hourly records from KSAT
Plume physics	576 steady-state solves on this site's own footprints; worst intake rise 0.1659 C at 40 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2022-03-25 (crossing)
Plant limit	21.0 C
Notice required	3 h
Forecast skill	0.50 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	2 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 11 of 24 hours with 2 mode changes. The reactive incumbent took 12 hours -- 1 more -- but declared 1 unsafe hour against the agent's 0. 1 hour was safe but still ran chillers, because the schedule could not afford them.

Agent	11 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
Incumbent	12 of 24 hours free cooling, 2 mode change(s), 1 unsafe hour(s) 1 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	FREE	yes	-	10.005	21.0	10.000	1.074
01	FREE	yes	-	9.723	21.0	9.440	0.929
02	FREE	yes	-	9.726	21.0	9.440	0.774
03	FREE	yes	-	8.359	21.0	6.670	0.643
04	FREE	yes	-	8.608	21.0	7.220	0.770
05	FREE	yes	-	8.333	21.0	6.670	0.691
06	FREE	yes	-	6.957	21.0	6.670	0.605
07	FREE	yes	-	7.352	21.0	6.110	0.583

08	FREE	yes	-	10.894	21.0	11.110	1.165
09	FREE	yes	-	15.708	21.0	16.804	2.170
10	FREE	yes	-	17.951	21.0	20.717	2.138
11	mechanical	no	dry-bulb	22.124	21.0	25.725	1.369
12	mechanical	no	dry-bulb	25.072	21.0	27.279	1.025
13	mechanical	no	dry-bulb	27.515	21.0	28.890	1.008
14	mechanical	no	dry-bulb	30.451	21.0	30.157	1.071
15	mechanical	no	dry-bulb	30.726	21.0	30.157	0.960
16	mechanical	no	dry-bulb	30.839	21.0	30.000	0.943
17	mechanical	no	dry-bulb	30.845	21.0	30.000	1.087
18	mechanical	no	dry-bulb	29.172	21.0	28.330	0.991
19	mechanical	no	dry-bulb	27.229	21.0	25.000	1.340
20	mechanical	no	dry-bulb	24.449	21.0	20.000	1.314
21	mechanical	no	dry-bulb	23.613	21.0	19.440	1.501
22	mechanical	no	dry-bulb	21.673	21.0	18.890	1.236
23	mechanical	yes	switch budget	18.343	21.0	17.220	0.975

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 10.005 C, which is 10.995 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.074 C, and the margin is measured, not chosen: 1.070 C of group-conditional forecast error for this hour of day, plus 0.0047 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

01:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 9.723 C, which is 11.277 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.929 C, and the margin is measured, not chosen: 0.926 C of group-conditional forecast error for this hour of day, plus 0.0031 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

02:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 9.726 C, which is 11.274 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.774 C, and the margin is measured, not chosen: 0.768 C of group-conditional forecast error for this hour of day, plus 0.0062 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

03:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 8.359 C, which is 12.641 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.643 C, and the margin is measured, not chosen: 0.632 C of group-conditional forecast error for this hour of day, plus 0.0119 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

04:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 8.608 C, which is 12.392 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.770 C, and the margin is measured, not chosen: 0.767 C of group-conditional forecast error for this hour of day, plus 0.0029 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

05:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 8.333 C, which is 12.667 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.691 C, and the margin is measured, not chosen: 0.688 C of group-conditional forecast error for this hour of day, plus 0.0029 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

06:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 6.956 C, which is 14.044 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.605 C, and the margin is measured, not chosen: 0.598 C of group-conditional forecast error for this hour of day, plus 0.0065 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

07:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 7.352 C, which is 13.648 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.583 C, and the margin is measured, not chosen: 0.570 C of group-conditional forecast error for this hour of day, plus 0.0133 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

08:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 10.894 C, which is 10.106 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.165 C, and the margin is measured, not chosen: 1.161 C of group-conditional forecast error for this hour of day, plus 0.0040 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

09:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.708 C, which is 5.292 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 2.170 C, and the margin is measured, not chosen: 2.156 C of group-conditional forecast error for this hour of day, plus 0.0139 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

10:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.951 C, which is 3.049 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 2.138 C, and the margin is measured, not chosen: 2.124 C of group-conditional forecast error for this hour of day, plus 0.0139 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.124 C against a 21.0 C limit, so it fails by 1.124 C. It would take a limit 1.124 C higher, or a bound 1.124 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.072 C against a 21.0 C limit, so it fails by 4.072 C. It would take a limit 4.072 C higher, or a bound 4.072 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 27.515 C against a 21.0 C limit, so it fails by 6.515 C. It would take a limit 6.515 C higher, or a bound 6.515 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 30.451 C against a 21.0 C limit, so it fails by 9.451 C. It would take a limit 9.451 C higher, or a bound 9.451 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 30.726 C against a 21.0 C limit, so it fails by 9.726 C. It would take a limit 9.726 C higher, or a bound 9.726 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 30.839 C against a 21.0 C limit, so it fails by 9.839 C. It would take a limit 9.839 C higher, or a bound 9.839 C tighter, to change this hour.

17:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 30.845 C against a 21.0 C limit, so it fails by 9.845 C. It would take a limit 9.845 C higher, or a bound 9.845 C tighter, to change this hour.

18:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 29.172 C against a 21.0 C limit, so it fails by 8.172 C. It would take a limit 8.172 C higher, or a bound 8.172 C tighter, to change this hour.

19:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 27.229 C against a 21.0 C limit, so it fails by 6.229 C. It would take a limit 6.229 C higher, or a bound 6.229 C tighter, to change this hour.

20:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.449 C against a 21.0 C limit, so it fails by 3.449 C. It would take a limit 3.449 C higher, or a bound 3.449 C tighter, to change this hour.

21:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.613 C against a 21.0 C limit, so it fails by 2.613 C. It would take a limit 2.613 C higher, or a bound 2.613 C tighter, to change this hour.

22:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.673 C against a 21.0 C limit, so it fails by 0.673 C. It would take a limit 0.673 C higher, or a bound 0.673 C tighter, to change this hour.

23:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 18.343 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

Generated by AGENTIC-ARBITER/src/report.py. Verified by being read back: the file was reopened after writing and every hour of the schedule, the headline counts and this site's own name were confirmed present.